

A Practical Guide to Stochastic gradient descent

TOPIC -- STOCHASTIC GRADIENT DESCENT

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$$

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Averaging Averaged stochastic gradient descent, invented independently by Ruppert and Polyak in the late 1980s, is ordinary stochastic gradient descent that records an average of its parameter vector over time. That is, the update is the same as for ordinary stochastic gradient descent, but the algorithm also keeps track of

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where ε is a small scalar (e.g. 10^{-8}) used to prevent division by 0, and β_1 and β_2 are the forgetting factors for gradients and second moments of gradients, respectively. Squaring and square-rooting is done element-wise. As the exponential moving averages of the gradient $m_w(t)$ and the squared gradient $v_w(t)$ are initialized with a vector of 0's, there would be a bias towards zero in the first training iterations. A factor $\frac{1}{1 - \beta_1^{1/2 t}}$ is introduced to compensate this bias and get better estimates $\hat{m}_w(t)$ and $\hat{v}_w(t)$. The initial proof establishing the convergence of Adam was incomplete, and subsequent analysis has revealed that Adam does not converge for all convex objectives. Despite this, Adam continues to be used due to its strong performance in practice.

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Backtracking line search Backtracking line search is another variant of gradient descent. All of the below are sourced from the mentioned link. It is based on a condition known as the Armijo–Goldstein condition. Both methods allow learning rates to change at each iteration; however, the manner of the change is different. Backtracking line search uses function evaluations to check Armijo's condition, and in principle the loop in the algorithm for determining the learning rates can

be long and unknown in advance. Adaptive SGD does not need a loop in determining learning rates. On the other hand, adaptive SGD does not guarantee the "descent property" – which Backtracking line search enjoys – which is that $f(x_{n+1}) \leq f(x_n)$ for all n . If the gradient of the cost function is globally Lipschitz continuous, with Lipschitz constant L , and learning rate is chosen of the order $1/L$, then the standard version of SGD is a special case of backtracking line search.

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Stochastic gradient descent (often abbreviated SGD) is an iterative method for optimizing an objective function with suitable smoothness properties (e.g. differentiable or subdifferentiable). It can be regarded as a stochastic approximation of gradient descent optimization, since it replaces the actual gradient (calculated from the entire data set) by an estimate thereof (calculated from a randomly selected subset of the data). Especially in high-dimensional optimization problems this reduces the very high computational burden, achieving faster iterations in exchange for a lower convergence rate. The basic idea behind stochastic approximation can be traced back to the Robbins–Monro algorithm of the 1950s. Today, stochastic gradient descent has become an important optimization method in machine learning.

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This equation is implicit since w^{new} appears on both sides of the equation. It is a stochastic form of the proximal gradient method since the update can also be written as:

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$$w^{\text{new}} := \arg \min_w \{ Q_i(w) + \frac{1}{2\eta} \|w - w^{\text{old}}\|^2 \}.$$

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Second-order methods A stochastic analogue of the standard (deterministic) Newton–Raphson algorithm (a "second-order" method) provides an asymptotically optimal or near-optimal form of iterative optimization in the setting of stochastic approximation. A method that uses direct measurements of the Hessian matrices of the summands in the empirical risk function was developed by Byrd, Hansen, Nocedal, and Singer. However, directly determining the required Hessian matrices for optimization may not be possible in practice. Practical and theoretically sound methods for second-order versions of SGD that

do not require direct Hessian information are given by Spall and others. (A less efficient method based on finite differences, instead of simultaneous perturbations, is given by Ruppert.) Another approach to the approximation Hessian matrix is replacing it with the Fisher information matrix, which transforms usual gradient to natural. These methods not requiring direct Hessian information are based on either values of the summands in the above empirical risk function or values of the gradients of the summands (i.e., the SGD inputs). In particular, second-order optimality is asymptotically achievable without direct calculation of the Hessian matrices of the summands in the empirical risk function. When the objective is a nonlinear least-squares loss

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Further proposals include the momentum method or the heavy ball method, which in ML context appeared in Rumelhart, Hinton and Williams' paper on backpropagation learning and borrowed the idea from Soviet mathematician Boris Polyak's 1964 article on solving functional equations. Stochastic gradient descent with momentum remembers the update Δw at each iteration, and determines the next update as a linear combination of the gradient and the previous update:

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$$w := w - \eta \nabla Q(w) = w - \eta \sum_{i=1}^n \nabla Q_i(w). \quad \{\displaystyle w:=w-\eta \, \nabla Q(w)=w-\frac{\eta}{n}\sum_{i=1}^n \nabla Q_i(w).\}$$

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As the algorithm sweeps through the training set, it performs the above update for each training sample. Several passes can be made over the training set until the algorithm converges. If this is done, the data can be shuffled for each pass to prevent cycles. Typical implementations may use an adaptive learning rate so that the algorithm converges. In pseudocode, stochastic gradient descent can be presented as :

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Notable applications Stochastic gradient descent is a popular algorithm for training a wide range of models in machine learning, including (linear) support vector machines, logistic regression (see, e.g., Vowpal Wabbit) and graphical models. When combined with the backpropagation algorithm, it is the de facto standard algorithm for training artificial neural networks. Its use has been also reported in the Geophysics community, specifically to applications of Full Waveform Inversion (FWI). Stochastic gradient descent competes with the

L-BFGS algorithm, which is also widely used. Stochastic gradient descent has been used since at least 1960 for training linear regression models, originally under the name ADALINE. Another stochastic gradient descent algorithm is the least mean squares (LMS) adaptive filter.